

SEMANTICQA

Revisiting *a Pain in the Neck*: A Semantic Reasoning Benchmark for Language Models

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| Motivation: A Blind Spot in LM Evaluation

Existing benchmarks emphasize math, code, and logic, but overlook **phrasal semantics**:

- **Sub-sentential meaning**: where meaning emerges from the interplay of lexical components and context.
- **Heuristic shortcuts**: even frontier LMs lean on surface heuristics rather than stable phrase-level representations.
- **Conflated formats**: existing tasks bundle multiple semantic operations, so strong scores may not reflect understanding.

We need diagnostic evaluation that **disentangles** semantic operations.

Interpretation

[Context] It was not *rocket science* in this case to determine that Goebbels was being cited to attack Trump, not to praise Nazis.

[Idiom] rocket science

[Interpretation] something complex and difficult

↕ Task Composition ↕

Extraction

[Context] It was not *rocket science* in this case to determine that Goebbels was being cited to attack Trump, not to praise Nazis.

[Extracted Idiom] rocket science

↕ Task Composition ↕

Classification

[Context] It was not *rocket science* in this case to determine that Goebbels was being cited to attack Trump, not to praise Nazis.

[Idiom] rocket science

[Choices]

(A) Projectile knowledge

(B) Difficult problem

(C) Proper Noun

(D) Meta Usage

✗

✓

✗

✗

Figure 1: Atomic task exemplars of idiomatic expression, grouped as task compositions.

| Research Question & Contributions

*How do language models behave when evaluated on phrasal semantics across **distinct** but **structurally constrained** task operations?*

SEMANTICQA contributes on three fronts:

1. **Operation-aligned Semantic Evaluation:** aligns existing SP tasks with the *semantic operations* they instantiate, enabling cross-operation analysis on the same phrase meaning.
2. **Minimal and Controlled Design:** fixed prompt templates hold structure constant while varying operations, supporting fair comparison under shared conditions.
3. **Diagnostic Analyses of Cascade Sensitivity:** sequential task compositions reveal phrase-level limitations that remain hidden in single-task evaluations.

SemanticQA: Framework Overview

SEMANTICQA benchmarks LMs on **4 phrase types** × **3 atomic operations**:

- **Phrase types:** Idiomatic Expressions (IE), Lexical Collocations (LC), Noun Compounds (NC), Verbal Constructions (VC).
- **Atomic operations:** *Classification* (multi-choice), *Extraction* (span identification), *Interpretation* (paraphrase generation).

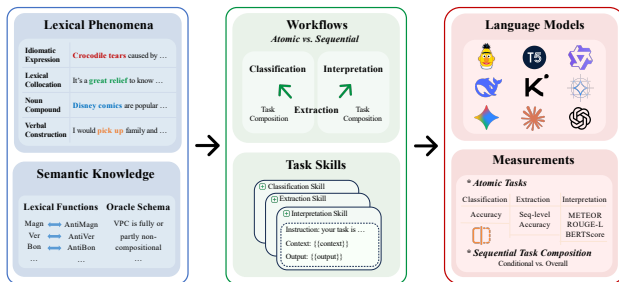


Figure 2: Overview of SEMANTICQA for benchmarking LMs on lexical phenomena.

| Phrase Type Taxonomy & Coverage

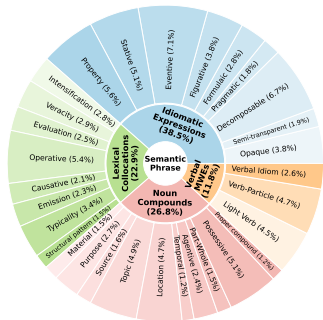


Figure 3: Coarse- and fine-grain semantic phrase categories of SEMANTICQA.

Four representative phrase types capture popular sources of variation:

- **LC**: base + collocate (e.g., heavy rain); compositional to idiom-style.
- **IE**: non-compositional (e.g., kick the bucket); needs conventionalized meaning.
- **NC**: compositional but ambiguous (e.g., baby oil vs. olive oil).
- **VC / VMwE**: VPC, LVC, VID; semi-compositional verbal expressions.

4 phrase types $\times$ 3 atomic operations $\Rightarrow$ **12 task variants**, spanning the full compositional spectrum.

Together, they expose **distinct** sources of semantic difficulty.

| Three Atomic Operations on the Same Phrase

Each phrase meaning is probed by **three structurally distinct operations** sharing a common context S :

- **Extraction:** identify the target span in S (strict structural constraint).
- **Classification:** pick the correct sense from candidate options.
- **Interpretation:** generate a contextualized paraphrase (open-ended).

Context: *"It was not rocket science in this case to determine that Goebbels was being cited..."*

Extraction: *rocket science* ✓

Classification: (A) Projectile knowledge ✗ **(B) Difficult problem** ✓ (C) Proper noun ✗

Interpretation: *"something complex and difficult"* ✓

| Benchmark Construction & Datasets

SEMANTICQA unifies prior resources into a single **operation-aligned** testbed:

- **IE:** detection, extraction, interpretation (273 / 447 / 818 examples).
- **LC:** categorization, extraction, interpretation (305 each, balanced by relation).
- **NC:** compositionality, extraction, interpretation (242 / 720 / 110).
- **VMwE:** VPC, LVC, VID extraction (475 examples total).

Phrase Type	Tasks	Output Constraint
Idiomatic Expressions	IED, IEE, IEI	Choice / Span / Paraphrase
Lexical Collocations	LCC, LCE, LCI	Choice / Span / Paraphrase
Noun Compounds	NCC, NCE, NCI	Choice / Span / Paraphrase
Verbal MwE	VPE, LVE, VIE	Span

Table 1: SEMANTICQA tasks. Datasets are deduplicated and reformatted for consistency across operations.

Overall Performance: The Capacity Triangle

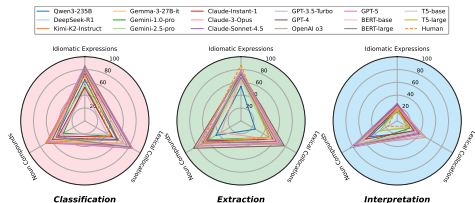


Figure 4: Best-performance capacity triangle (Δ) of LMs on SEMANTICQA.

Axes: Extraction (Acc_s) | Classification (Acc) | Interpretation (MTR); each axis aggregates over 4 phrase types.

Each axis is one atomic operation. A balanced model fills the triangle:

- **No model dominates uniformly:** strengths vary by operation *and* phrase type.
- **Operation-specific:** each operation imposes distinct structural constraints.
- **Diagnostic, not a leaderboard:** SEMANTICQA is neither saturated nor uniformly difficult.

Triangle area indexes **breadth**; vertices reveal **operation-specific** weak spots.

| Main Results: LMs Struggle Differently Across Operations

Model	Idiom (IE)			Collocation (LC)			Noun Comp. (NC)			VMwE
	IED	IEE	IEI	LCC	LCE	LCI	NCC	NCE	NCI	avg.
HUMAN	71.0	87.0	20.5	47.0	50.0	16.7	71.0	73.0	17.2	72.7
DEEPSEEK-R1 (5-shot)	84.3	72.3	19.2	76.1	64.3	32.9	60.6	70.7	68.7	58.2
KIMI-K2-INST. (5-shot)	81.7	69.6	21.7	79.7	69.2	36.9	64.7	63.6	76.7	57.0
CLAUDE-SONNET-4.5 (5-shot)	78.0	72.0	26.7	76.1	72.7	40.8	70.1	62.1	83.8	55.6
OPENAI O3 (5-shot)	83.5	74.7	21.9	83.6	71.5	35.9	63.5	78.6	74.5	54.7
GPT-5 (5-shot)	85.4	78.7	22.5	84.3	68.9	37.4	67.2	79.0	75.3	54.5

Table 2: Major results. Acc / Acc_S / MTR depending on task. Even frontier LMs trail humans on extraction and interpretation.

- **No single winner:** different LMs lead on different operations and phrase types.
- **Persistent gap on interpretation:** MTR remains low (e.g., IEI < 30%) despite strong B-S, indicating fluency outpaces grounding.

Cross-Task Variance: Mean vs. Std by Category

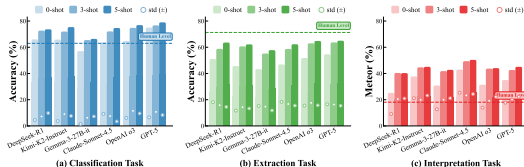


Figure 5: Bars = mean per category; markers = population std across tasks.

Read this as: bar height = *average* score; marker spread = *volatility* across the 12 tasks. Tall + tight $\Rightarrow$ broadly competent; tall + wide $\Rightarrow$ uneven.

What the variance reveals:

- **High std** on extraction signals *operation-specific* brittleness, not uniform difficulty.
- **Interpretation:** compressed mean but moderate std under flexible metrics.
- Strong mean $\neq$ consistent competence; **cross-task spread matters**.

Mean reports **average** skill; std reports **stability** — both are needed.

Semantic Category Scaling with In-Context Learning

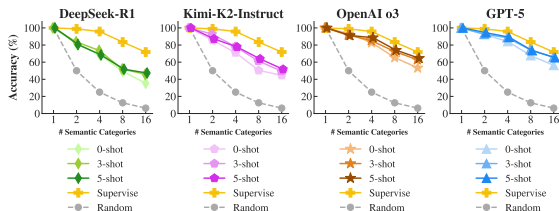


Figure 6: LC categorization with $n \in \{1, 2, 4, 8, 16\}$ classes and $k \in \{0, 3, 5\}$ shots.

Larger n exposes a steeper decay; more shots help only marginally.

Findings on category scaling:

- All LMs beat random / majority baselines $\Rightarrow$ non-trivial semantic reasoning even zero-shot.
- Accuracy **drops sharply** as n grows: DEEPSEEK-R1 81.7% $\rightarrow$ 35.4%.
- Supervised baselines degrade more gracefully than frontier LMs.

ICL alone does not substitute for explicit supervision under fine-grained relational distinctions.

| Sequential Task Compositions: Where Pipelines Break

Realistic workflows chain operations: *extract* $\rightarrow$ *interpret* or *extract* $\rightarrow$ *classify*.

- **Cond. MTR** (on correctly-extracted phrases) is consistently **much higher** than **Overall MTR** (end-to-end).
- Interpretation *cannot rescue* upstream extraction errors $\Rightarrow$ structural errors propagate.
- Compositional classification degrades sharply as relational granularity grows (4 $\rightarrow$ 8 $\rightarrow$ 16 classes).

Type	Model	Ext. (Acc)	Cond. (MTR)	Overall (MTR)
LC	DEEPSEEK-R1 (5-shot)	33.8	42.3	14.3
	GPT-5 (5-shot)	41.3	41.8	17.3
IE	DEEPSEEK-R1 (5-shot)	57.0	13.4	7.6
	GPT-5 (5-shot)	59.3	17.1	10.1

Table 3: Sequential extraction $\rightarrow$ interpretation. Cond. MTR $\gg$ Overall MTR exposes the extraction bottleneck.

| VMwE Extraction with Oracle Schema

Oracle Schema: augment the prompt with the *target VMwE type* and its definition (VPC / LVC / VID).

- Consistently lifts performance across frontier LMs.
- DEEPSEEK-R1: **51.6%** → **64.1%** on VMwE extraction.
- Explicit semantic descriptions of the target structure substantially aid identification.

Lesson: *structured task instructions narrow the gap when phrase-type cues are otherwise ambiguous.*

| Discussion & Takeaways

Three lessons from operation-aligned semantic evaluation:

1. **Phrasal Semantics Requires Multi-Dimensional Evaluation.** A single task or metric cannot capture phrase-level competence; extraction, classification, and interpretation probe complementary aspects.
2. **Metric Sensitivity Shapes Apparent Strengths.** Strong B-S on interpretation primarily reflects paraphrase fluency; strict span metrics expose brittleness in structural identification. *High interpretation scores $\nRightarrow$ semantic correctness.*
3. **Workflow Robustness Remains Limited.** Sequential pipelines are highly sensitive to upstream errors; interpretation does not compensate for failed extraction. Cascade fragility stays hidden under flexible metrics.

| Conclusion

- **SEMANTICQA**: an *operation-aligned* benchmark that decomposes phrasal semantics into classification, extraction, and interpretation across IE, LC, NC, and VMwE.
- **Diagnostic, not saturated**: frontier LMs (GPT-5, OpenAI o3, Claude-Sonnet-4.5, DeepSeek-R1, and Kimi-K2) all leave headroom; no model dominates every operation.
- **Cascade-aware**: sequential compositions reveal extraction as the dominant bottleneck for end-to-end semantic pipelines.

Thank You !



Project Page: <https://semanticqa.github.io>



Code & Data: <https://github.com/jacklanda/SemanticQA>



Datasets: <https://huggingface.co/datasets/jacklanda/SemanticQA>